Keuntae Kim

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Summary

PhD Candidate at Hanyang University working across three areas. **AI Safety & Alignment**: how hidden traits travel into models through RL rewards and distillation data, why banning their surface channels only moves them elsewhere, and where they can be cut — down to single heads and neurons; plus white-box detection of hallucination and reasoning quality. **LLM / VLM Post-training**: multi-rubric RLAIF and RLVR, agentic RL for stateful agents, and continual learning that keeps whole networks learning without replay, distillation or freezing. **Multimodal & Embodied Models**: Video-LLMs and world dynamics, reasoning in diffusion MLLMs, and test-time embodiment alignment for vision-language-action (VLA) models. This has produced **5 first-author papers** at **NeurIPS 2025 / CVPR 2026 / ICML 2026 / EMNLP 2026 / NeurIPS 2026** and **7** first-author submissions under review at ICLR 2027 and Nature (three at Nature, each with a public preprint and code), plus a VLA preprint with public code. I also own the **distributed training stack** end-to-end, training LLMs and VLMs on **up to 32 H100s (4 nodes × 8)** across Samsung SDS Cloud, NHN Cloud (B200), and AWS (H200). **Researcher** on two Korean national AI projects (NRF, IITP); reviewer for NeurIPS, ICLR, ICML, CVPR, ECCV, and ACL ARR.

Experiences

Large-Scale & Multi-Node Training Experience *Samsung SDS Cloud / NHN Cloud / AWS*

Hands-on owner of end-to-end distributed training for LLMs and VLMs on three production GPU clouds — cluster bring-up, NCCL / RDMA validation, parallelism strategy, and fault-tolerant multi-day runs.

Platform	Accelerator	Topology	Scale
Samsung SDS Cloud	NVIDIA H100 SXM (80 GB)	4 nodes × 8 GPUs	**32 GPUs** — largest configuration; NVLink/NVSwitch intra-node, RDMA inter-node
NHN Cloud	NVIDIA B200 (Blackwell)	2 nodes × 8 GPUs	**16 GPUs** — next-gen Blackwell bring-up, FP8/BF16 mixed-precision training
AWS	NVIDIA H200 SXM (141 GB)	2 nodes × 8 GPUs	**16 GPUs** — EFA/RDMA multi-node fabric, HBM3e-enabled long-context training

- **Parallelism & memory.** Trained models beyond single-node memory with DeepSpeed ZeRO-3, PyTorch FSDP, and Megatron tensor / pipeline parallelism, combined with activation checkpointing, Liger kernels, `torch.compile`, and FlashAttention-2/3 under BF16 / FP8 mixed precision.
- **Workloads.** Full-parameter and LoRA fine-tuning of **7B–30B** LLMs via **ms-swift** and **LLaMA-Factory**; VLM / Video-LLM multimodal training (vision encoder + projector + LLM stages); and **RLVR / policy optimization** (GRPO, PPO) on **verl**, **NeMo-RL**, and **slime** with vLLM-backed distributed rollouts.

AI Safety & Alignment

Subliminal RL — Trait Transfer via LLM-Judge Rewards *Nature (under review)* arXiv kkt94/subliminal-rl

- Showed that **GRPO with an LLM judge is enough for subliminal learning**: a student that only writes invented syllables or numbers, scored by a judge with a hidden preference, moves its own preference toward the judge's trait relative to a matched run under a judge with another trait — no teacher text ever enters the update.
- Ran a **48-cell** design grid (prompted vs. fine-tuned judges, syllable vs. numbers-only carriers): **17 cells** establish the trait at 5% FDR, numbers-only cells included; **removing only the reward difference between judges removes transfer**.
- Established the principle as a **four-stage chain** — emission, absorption, transfer, behaviour — and carried it unchanged to **AI-safety dispositions**: undesirable dispositions reach the student's answers, and stages measured *before* training anticipate where.

Balloon Effect — Hidden Traits under Token Bans *Nature (under review)* arXiv kkt94/balloon-effect

- Stress-tested the intuitive defence against subliminal learning — banning the tokens through which a trait shows — across **2 models** (Qwen3-4B, Llama-3.2-3B), **6 word-free output formats** and **16 traits**: banning the most over-represented feature produced a new one in **102/192** cells (0 of 24 null runs); banning a format's whole vocabulary moved the trait into its name, and banning the name into its fragments — the **balloon effect**.

- Traced the trait **down to single heads and neurons**: a small attention circuit in prompted teachers and a key–value memory written by the loss in fine-tuned teachers.
- Patching that circuit while the teacher writes ordinary numbers stopped the chains, sharply reduced the overt name leak and **halved** the trait passed to a student, whereas as many random heads produced no detectable reduction.

CLAWS — Creativity Detection for LLM Solutions *NeurIPS 2025* kkt94/CLAWS

- Proposed **CLAWS**, a white-box detector using attention across prompt sections vs. output to classify solutions as *Typical / Creative / Hallucinated* — the first three-way creativity assessment for reasoning, with no human evaluation in the loop.
- Validated on **5 7–8B math RL models** over **4,545 problems from 181 contests** (AJHSME, AMC, AIME), outperforming **5 white-box baselines** (Perplexity, Logit/Window Entropy, Hidden Score, Attention Score) on both creativity and hallucination detection.
- Ensured evaluation rigor with **5 evaluation strategies** (threshold, prototype, MLP, XGBoost, TabM) and macro-averaged metrics for the imbalanced Creative / Hallucinated classes.

LLM / VLM Post-training (RL, Continual Learning)

Heat-GRPO — Heat-Kernel Policy Optimization for Multi-Rubric RL *ICLR 2027 (under review)*

- Built a **multi-rubric RL (RLAIF) training pipeline** in **verl** (FSDP + async vLLM rollouts) with a served **Qwen3-32B LLM-judge** returning per-rubric YES/NO log-prob margins as rewards — 8 rollouts per prompt, up to 16 signed rubrics per example.
- Implemented and trained **GRPO, GDPO, GD^{2}PO**, and my **Heat-GRPO** (Gaussian-CDF scoring of judge margins before aggregation) on **6 policies** (Llama 1B–8B, Qwen3 1.7B–8B) × **4 benchmarks** (RaR-Science/Medicine, BeaverTails, HealthBench) × 3 seeds on H100s — **288 full RL runs**.
- Heat-GRPO achieved the best mean in **23/24** conditions (up to **+25%**), held up under **3 additional judges**, and cut advantage-computation cost by **∼39%** vs. GDPO / GD^{2}PO.

What Gets Deployed? — Deployment Accounting for Stateful Agents *ICLR 2027 (under review)*

- Trained **LLM agents with multi-turn RL** in interactive environments (**ALFWorld**, **WebShop**) using **verl-agent + Ray**: Qwen2.5-7B, full-parameter FSDP on 4×H100, 16 sandboxed environment rollouts per task, 50-step action horizons.
- Implemented and compared **10 agentic RL credit-assignment variants** — GRPO, GiGPO (step-level credit), PKPO-style pass@k, leave-one-out multi-sample utilities, ternary abstention rewards, and a commit-aware credit — plus exploratory AppWorld runs (Qwen3.5-9B, LoRA).
- Built a **deployment-accounting** evaluation (commit / wrong-commit / abstain over sampled executions): trained policies reached **72.5% / 71.3%** committed success on seen / unseen rooms vs. 62.9% / 60.5% for GRPO, with outcome selection adding **9–13 points**.

DeltaHippo — Replay-Free Hippocampal Consolidation *Nature (under review)* arXiv kkt94/deltahippo

- Built a **training-only, hippocampus-inspired memory** that lets a **fully fine-tuned** network keep learning: a gradient-free "sleep" after each task indexes every synapse group's input geometry (gated-delta write/erase), then constrains the gradient *and* the AdamW update, with state-only conflict detection for near-duplicate classes — **no replay, teacher, frozen modules, or inference-time memory**.
- Ran full fine-tuning of **10 models across 4 families** (Qwen3 / Llama-3 LLMs up to **8B**, BERT, RoBERTa, ViT-B/16, ResNet-50) on **6 class-incremental benchmarks**, 3 seeds each, on H100 and B200 — the first single principle shown across all four families.
- Beat sequential fine-tuning (+25–77 points), subspace-projection methods, replay, and distillation on every cell; led the frozen-backbone baseline by **+17.5 points** on TACRED and cut near-duplicate absorption to **2.8%** vs. 9–41% for every compared method.

Revisiting Continual Learning — Three-Interface Diagnosis *NeurIPS 2026* kkt94/RevisitingCL

- Showed that **BWT and FWT are interface-dependent**: the sequential head suggests catastrophic forgetting, a fresh probe near-complete recoverability, and a classifier-free backbone diagnostic only **partial drift**.

- Found a **backbone-level stability–plasticity trade-off** that is negative in **all 150** clustering configurations (mean $r=-0.74$) while head and probe correlations stay mixed — across **7 datasets**, encoder and decoder backbones up to 8B, and 8+ IL methods.
- Introduced **Just LM-Head Tuning (JLT)**, a head-only diagnostic intervention on the fixed incrementally trained backbone that lifts accuracy by **+53 to +64 points** and recovers **96–99%** of the gap to joint training for replay- and distillation-based methods (e.g., 40.7 → 95.6 vs. 96.4 joint) but only about three quarters for plain sequential fine-tuning — recasting much of "catastrophic forgetting" as **head–backbone mismatch** plus partial drift.

Multimodal & Embodied Models (Video-LLM, Diffusion MLLM, VLA)

TEA — Self-Calibrating VLAs *Preprint* kkt94/self-calibrating-vla

- Showed that much of the failure of vision-language-action (VLA) models under a **perturbed robot initial state** or a **moved camera** is not in the policy: the embodiment departs from the one the VLA was trained on, and the VLA cannot tell.
- Proposed **Test-Time Embodiment Alignment (TEA)**: forward models predict how the training embodiment would realize each command and how the robot should appear to its camera, and correct the VLA's commands (*State Alignment*) and view (*View Alignment*) only when the evidence shows a departure — policy untouched, controller-agnostic, no data from the deployment condition.
- Across **5 benchmarks, 3 robots and 6 VLAs**: up to **+59 points** initial-state and **+21 points** camera-perturbed success on LIBERO-Plus, persistent action bias recovered from 38–60% to **81–88%**, carrying over to a dual-arm robot and the Google Robot in SimplerEnv with unperturbed performance intact.

Projector-JEPA — Two-Expert Residual Projector for Video-LLMs *ICLR 2027 (under review)*

- Proposed a **two-expert residual projector** (Local + State Expert, both Gated DeltaNet) on any frozen Video-LLM, specialized by **JEPA-style objectives** (local correspondence + masked state reconstruction) to model inter-frame motion and world dynamics.
- Trained only an **∼80K-parameter** module for a **single epoch on 70K public samples**, yielding **up to ∼7%** gains across **7 Video-LLMs** (PLM, VideoLLaMA3, InternVL3 families) on **5 benchmarks** (MVBench, TempCompass, NExT-QA, WorldBench, Perception Test).

Thinking Diffusion & ST-Veto — Diffusion MLLM Reasoning ***CVPR 2026*** | ***ICML 2026***

- **Thinking Diffusion**: first quantitative analysis of CoT in dMLLMs, identifying *early answer generation* and *weak early-step visual grounding*; training-free **PSP** and **VRG** give **up to +7.5%** accuracy at **>3×** faster inference on LaViDa and MMaDA.
- **ST-Veto**: a decoding-time **veto-and-swap** that replaces temporally unstable tokens (second-order Taylor prediction of confidence) and weakly grounded tokens (image-attention mass) before unmasking — **up to +9%** with **zero extra training or generation cost**.

Education

Hanyang University, Ph.D. in Computer Science — Sep. 2023 – Present
Advisor: Prof. Yong Suk Choi | Seoul, Korea

Hanyang University, M.S. in Computer Science — Sep. 2021 – Aug. 2023
Advisor: Prof. Yong Suk Choi | Seoul, Korea

Kookmin University, B.S. in Automobile and IT Convergence — Mar. 2015 – Aug. 2019
Seoul, Korea

Research Interests

- **AI Safety & Alignment** — hidden-trait transfer (subliminal learning) in RL and distillation, mechanistic interpretability, white-box detection of hallucination and reasoning quality
- **LLM / VLM Post-training** — RLVR and multi-rubric RLAIF, agentic RL for stateful agents, continual learning (stability–plasticity)

- **Multimodal & Embodied Models** — Video-LLMs and world dynamics, diffusion MLLMs and visually grounded reasoning, vision-language-action (VLA) models

Publications (Journals & Conferences)

Bold + underline = *author;* ⋆ = *first author.*

1. **Keuntae Kim**⋆, Yong Suk Choi. "CFP-AL: Combining Model Features and Prediction for Active Learning in Sentence Classification." *Applied Sciences*, 2025. PUBLISHED [paper]
2. **Keuntae Kim**⋆, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. "CLAWS: Creativity detection for LLM-generated solutions using Attention Window of Sections." *NeurIPS 2025*. PUBLISHED [arXiv:2510.17921] [code]
3. **Keuntae Kim**⋆, Mingyu Kang, Yong Suk Choi. "Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models." *CVPR 2026*. PUBLISHED [arXiv:2604.05497]
4. **Keuntae Kim**⋆, Beomseok Lee⋆, Hyunwoo Kim⋆, Yong Suk Choi. "ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding." *ICML 2026*. PUBLISHED [arXiv:2607.17884]
5. **Keuntae Kim**⋆, Eunhye Jeong, Yong Suk Choi. "CWM: Controllable White-Box Meta-Prompting for Adaptive Retrieval-Augmented Generation and Reasoning Ability." *Findings of EMNLP 2026*. PUBLISHED [arXiv:2609.15234]
6. **Keuntae Kim**⋆, Beomseok Lee⋆, Hyunwoo Kim⋆, Yong Suk Choi. "Revisiting Continual Learning in Language Models: A Three-Interface Diagnosis of Stability and Plasticity." *NeurIPS 2026*. PUBLISHED [paper] [code]
7. **Keuntae Kim**⋆, Yong Suk Choi. "Two Clocks in Diffusion MLLMs: Answer Stabilization and Rationale Writing." *NeurIPS 2026 Workshop (BeNTo)*. PUBLISHED
8. **Keuntae Kim**⋆, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. "Heat-GRPO: Heat-Kernel Group Relative Policy Optimization for Multi-Rubric Reinforcement Learning." *ICLR 2027 (under review)*. UNDER REVIEW
9. **Keuntae Kim**⋆, JungHyoun Kim, Dongjun Kim, HyeongSeok Kang, Yong Suk Choi. "Projector-JEPA: A Gated DeltaNet Two-Expert Residual Projector Bridging Video-LLMs to World-Dynamics Understanding." *ICLR 2027 (under review)*. UNDER REVIEW
10. **Keuntae Kim**⋆, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. "Can RLVR Models Think More Creatively? A Study of Overconfidence and High Entropy Segment." *ICLR 2027 (under review)*. UNDER REVIEW
11. **Keuntae Kim**⋆, Yong Suk Choi. "Subliminal Reinforcement Learning: Traits Travel Through a Judge's Scores." *Nature (under review)*. UNDER REVIEW [arXiv:2610.41301]
12. **Keuntae Kim**⋆, Yong Suk Choi. "A Balloon Effect: Ban One Channel and a Hidden Trait Finds Another." *Nature (under review)*. UNDER REVIEW [arXiv:2611.02509]
13. **Keuntae Kim**⋆, Yong Suk Choi. "Hippocampal Consolidation without Replay in Fully Fine-tuned Networks." *Nature (under review)*. UNDER REVIEW [arXiv:2611.01952]
14. Beomseok Lee⋆, **Keuntae Kim**⋆, Hyunwoo Kim, Yong Suk Choi. "What Gets Deployed? Deployment Accounting for Stateful Agents." *ICLR 2027 (under review)*. UNDER REVIEW
15. JungHyoun Kim, **Keuntae Kim**, HyeongSeok Kang, Dongjun Kim, Yong Suk Choi. "Fourier-World: Temporal Fourier States for Visual World Models." *ICLR 2027 (under review)*. UNDER REVIEW
16. Dongjun Kim, **Keuntae Kim**, JungHyoun Kim, HyeongSeok Kang, Yong Suk Choi. "Complete the Scene, Cover the Video: Recurrent Token Compression for Video LLMs." *ICLR 2027 (under review)*. UNDER REVIEW
17. **Keuntae Kim**⋆, Yong Suk Choi. "Self-Calibrating VLAs: Test-Time Embodiment Alignment for Vision-Language-Action Models." PREPRINT [paper]

Research Grants & Funded Projects

- **Metacognitive AGI Framework and its Applications** — IITP. *Lead Student Researcher* (Jul. 2025 – Present).
- **Metacognitive Generative AI Framework for Mitigating Hallucination in LLMs** — NRF. *Lead Student Researcher* (Feb. 2025 – Present).
- **Big Data–based Real-time Monitoring and Social Simulation System** — NRF. *Lead Student Researcher, Group 3* (Sep. 2021 – Feb. 2025).
- **High-performance Deep Learning via Cross-connection of Autoencoders and Decoders** — NRF. *Researcher* (Sep. 2021 – Feb. 2024).
- **Natural Language Data Refinement for Large-scale LM Construction** — KT Corp. *Researcher* (Sep. 2021 – Jul. 2022).

Patents (Korea)

- "Fusion of Sparse-Dense and Global Attention for Video Action Recognition," KR 10-2024-0132801.
- "Device and Method for Correcting Toxic Words in Sentence," KR 10-2024-0143175.
- "Method and Device for Displaying Emotion Map Animation," KR 10-2024-0152245.
- "Method for Analysing Emotion Based on SNS Data," KR 10-2023-0193546.

Technical Skills

- **Training** — PyTorch, DeepSpeed (ZeRO-2/3), PyTorch FSDP, Megatron-LM, ms-swift, LLaMA-Factory, HF Accelerate, PEFT/LoRA, Liger kernels, bitsandbytes
- **RL & post-training** — verl, verl-agent, NeMo-RL, slime, TRL; GRPO / PPO / RLVR / multi-turn agentic RL with vLLM rollouts
- **Distributed & HPC** — multi-node training (up to 32×H100 / 4 nodes), SLURM, Ray, NCCL over InfiniBand / RoCE / AWS EFA, torchrun, tensor / pipeline parallelism, FlashAttention-2/3, BF16 & FP8, Megatron-Bridge / checkpoint-engine
- **Data & evaluation** — Data-Juicer, NeMo Curator, lm-evaluation-harness, OpenCompass, EvalScope
- **GPU platforms** — NVIDIA H100 / H200 / B200 (Blackwell), A100; Samsung SDS Cloud, NHN Cloud, AWS
- **Serving & inference** — vLLM, SGLang, TensorRT-LLM, HF TGI, ONNX Runtime, llama.cpp
- **Languages** — Python, C/C++, CUDA, JavaScript, TypeScript, Java, SQL
- **Web & backend** — Flask, FastAPI, Django, Node.js, React, REST APIs, WebSocket
- **Infra** — Linux, Docker, Git, W&B, MLflow, TensorBoard, NVIDIA Nsight / PyTorch Profiler

Awards & Honors

- **NeurIPS 2025 Scholar Award**
- **BK21 FOUR Scholarship** — Korean government-funded research fellowship, 2021 – Present

Academic Service & Activities

Conference Reviewer

- *Machine Learning* — **NeurIPS** 2026 | **ICLR** 2027 | **ICML** 2026
- *Computer Vision* — **CVPR** 2026 | **ECCV** 2026
- *Natural Language Processing* — **ACL Rolling Review** 2026.01 – Present

Workshop Presentation

- **NeurIPS 2026** BeNTo Workshop — *Two Clocks in Diffusion MLLMs*